

Winter 2024 Computer Networks & Security (SFWR ENG 4C03)

Instructor: Rong Zheng

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Lecture time: MoTh 9:30AM - 10:20AM, Tu 10:30AM - 11:20AM

Class Room: CNH 104

Tutorial: We 11:30AM - 12:20PM (ABB 163, JHE 326H), Mo 11:30AM - 12:20PM (ABB 164), Th 11:30AM - 12:20PM (JHE 326H)

Office Hours: Tue. 5 – 6pm

Avenue: <http://avenue.mcmaster.ca/>

TAs:

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Textbook:

- James F. Kurose, Keith W. Ross, "Computer Networking: A Top-Down Approach Featuring the Internet", 8th ed. Pearson Education

Synopsis:

Computer networking is a rapidly advancing field. The Internet is already an integral part of society. It is important for computer scientists and computer engineers to be familiar with the fundamentals as well as practices of computer networking. This graduate course will emphasize on the algorithms, protocols and performance evaluation of the Internet. Topics include TCP/IP protocol stack, routing, congestion control, wireless networks and selected materials in network security.

Grading:

- Online quizzes (~5): 25% (*typically due in one week*)
- 5 Assignments: 35% total
- Final: 40%
- **3% bonus** for top 15% students who answer the most questions and/or ask the most GOOD questions on Avenue
- **3% bonus of classroom participation:** for top 15% students who ask meaningful questions or answer questions in class

MSAF policy:

- Missed quizzes with MSAF approvals will be counted toward the total quiz grade
- Missed assignments with MSAF approval need to be completed in 10 days from the due day; otherwise, starting from the 11th day, the late submission policy applies.
- The list of approved MSAFs will be shared with students in the last week of class

Late submission policy:

- a deduction of 10% of the maximum mark available from the actual mark achieved by the student shall be imposed upon expiry of the deadline;
- a further deduction of 5% of the maximum mark available from the actual mark achieved by the student shall then be imposed on each of the next subsequent days;

Academic Dishonesty

You are expected to exhibit honesty and use ethical behavior in all aspects of the learning process. Academic credentials you earn are rooted in principles of honesty and academic integrity.

Academic dishonesty is to knowingly act or fail to act in a way that results or could result in unearned academic credit or advantage. This behaviour can result in serious consequences, e.g. the grade of zero on an assignment, loss of credit with a notation on the transcript (notation reads: "Grade of F assigned for academic dishonesty"), and/or suspension or expulsion from the university.

It is your responsibility to understand what constitutes academic dishonesty. For information on the various types academic dishonesty please refer to the Academic Integrity Policy, located at

<http://www.mcmaster.ca/academicintegrity>

The following illustrates only three forms of academic dishonesty:

- Plagiarism, e.g. the submission of work that is not one's own or for which other credits have been obtained.
- Improper collaboration in group work.
- Copying or using unauthorized aids in tests and examinations.

MOSS will be used for plagiarism detection for programming assignments in this course.

Faculty Notices

"The Faculty of Engineering is concerned with ensuring an environment that is free of all discrimination. If there is a problem, individuals are reminded that they should contact the Department Chair, the Sexual Harassment Officer or the Human Rights Consultant, as the problem occurs."

Tentative Schedule:

Lecture	Topic	Reading	Assignments	Tutorial	TA
Week 1 (Jan. 6 – Jan. 10)	Course overview Evolution of the Internet	[Kur05] Chapter 1			
Week 2 (Jan. 13 – Jan. 17)	Fundamental of the Internet Application Layer	[Kur05] Chapter 1 - 2	Q1 (5pt)	Wireshark walkthrough	
Week 3 (Jan. 20 – Jan. 24)	Application Layer	[Kur05] Chapter 2	Q2 (5pt)	Q&A	
Week 4 (Jan. 27 – Jan. 31)	Transport Layer	[Kur05] Chapter 3	A1: Wireshark (5pt)	Socket programming examples	
Week 5 (Feb. 3 – Feb. 7)			Q3 (5pt)	Q&A	
Week 6 (Feb. 10 – Feb. 14)	Routing	[Kur05] Chapter 4	A2: Socket Programming (10pt)	More on socket programming	
Week 7: Midterm recess					
Week 8 (Feb. 24 – Mar. 28)	Routing	[Kur05] Chapter 4	Q4 (5pt)	Routing algorithm examples	
Week 9 (Mar. 3 – Mar. 7)	Link layer	[Kur05] Chapter 5	A3: Wireshark (5pt)	Q&A	
Week 10 (Mar. 10 – Mar. 14)	Wireless Networks	[Kur05] Chapter 6	A4: Mininet I (5pt)	Linux, VM, network emulation	
Week 11 (Mar. 17 – Mar. 21)			Q5 (5pt)	Q&A	
Week 12 (Mar. 24 – Mar. 28)	Network Security	[Kur05] Chapter 8	A5: Mininet II (10pt)	MAC and self-learning algorithm examples	
Week 13 (Mar. 31– Apr. 4)			Q6 (5pt, used to substitute the lowest quiz score)	Q&A	
Week 13 (Apr. 7)	Final review			Q&A	

